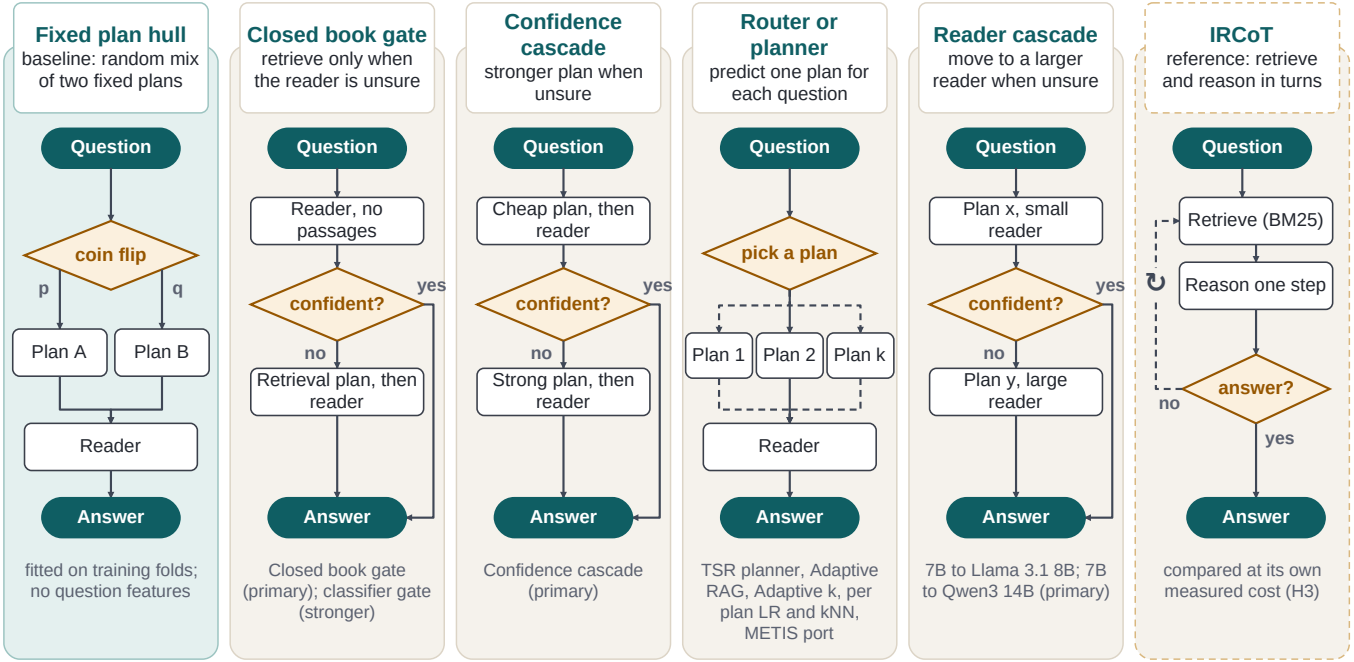
Fixed plan hull
baseline: random mix of two fixed plans
Question
coin flip
p
q
Plan A
Plan B
Reader
Answer
fitted on training folds; no question features
Closed book gate
retrieve only when the reader is unsure
Question
Reader, no passages
confident?
yes
no
Retrieval plan, then reader
Answer
Closed book gate (primary); classifier gate (stronger)
Confidence cascade
stronger plan when unsure
Question
Cheap plan, then reader
confident?
yes
no
Strong plan, then reader
Answer
Confidence cascade (primary)
Router or planner
predict one plan for each question
Question
pick a plan
Plan 1
Plan 2
Plan k
Reader
Answer
TSR planner, Adaptive RAG, Adaptive k, per plan LR and kNN, METIS port
Reader cascade
move to a larger reader when unsure
Question
Plan x, small reader
confident?
yes
no
Plan y, large reader
Answer
7B to Llama 3.1 8B; 7B to Qwen3 14B (primary)
IRCoT
reference: retrieve and reason in turns
Question
Retrieve (BM25)
Reason one step
answer?
no
yes
Answer
compared at its own measured cost (H3)